

SECP3744 - UTHM CASE STUDY
UTHM INVENTORY SYSTEM PROJECT REPORT
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UTHM CASE STUDY
UTHM INVENTORY SYSTEM
INTEGRATED INVENTORY MANAGEMENT SYSTEM FOR UTHM

TAN ZHI MING

A project report submitted in elearning of the
requirements for the award of the degree of
Bachelor of Data Engineering

School of Computer Science (Data Engineering)
Faculty of Computing
Universiti Teknologi Malaysia

DECEMBER 2025

DECLARATION

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Date : 12 JANUARY 2026

ACKNOWLEDGEMENT

In preparing this thesis, I was in contact with many people, researchers, academicians, and practitioners. They have contributed towards my understanding and thoughts. In particular, I wish to express my sincere appreciation to my course lecturer for Enterprise Systems Design and Modeling (WBL), Dr. Aryati binti Bakri, for her encouragement, continuous guidance, and constructive critics throughout the semester.

Her deep knowledge of Enterprise Architecture and System Integration provided the crucial foundation for the completion of this UTHM Inventory System case study. Without her continued support and interest, this project report would not have been the same as presented here.

I am also indebted to Universiti Teknologi Malaysia (UTM) for providing the platform for this study. My sincere appreciation also extends to all my fellow teammates who have worked tirelessly to ensure the success of this project. Their views, cooperation, and tips were indeed useful.

Finally, I am grateful to all my family members for their endless support and understanding during the completion of this project.

ABSTRACT

University Tun Hussein Onn Malaysia (UTHM) aims to achieve full digital transformation by 2030, yet its current asset management process faced bottleneck by legacy client-server system and manual dependencies. This limitation created Information Silos between the Bursary Department and Faculties, leading to data fragmentation, accessibility for outstation staff, and inefficient for maintenance and tracking item. This project solve these challenges by build a inventory subsystem. The system utilizing the System Development Life Cycle methodology and The Open Group Architecture Framework (TOGAF) for Enterprise Architecture analysis. The proposed solution will help UTHM transit from a rigid legacy environment to a scalable, web based Three-Tier architecture hosted within the university Data Center. The system key contribution include a real time centralized database, a mobile-responsive interface for accessibility and a specialized Repair Module that automate report and blocking of defective asset via QR code scanning. The final deisgn provides a blueprint for a unified enterprise system that enhanced user experience, ensures regalatory compliance and support UTHM in operational excellence.

ABSTRAK

Universiti Tun Hussein Onn Malaysia (UTHM) menyasarkan untuk mencapai transformasi digital sepenuhnya menjelang 2030, namun proses pengurusan aset semasa menghadapi kesesakan oleh sistem pelayan pelanggan lama dan kebergantungan manual. Batasan ini mewujudkan Silo Maklumat antara Jabatan Bendahari dan Fakulti, yang membawa kepada pemecahan data, kebolehcapaian untuk kakitangan stesen luar, dan tidak cekap untuk penyelenggaraan dan item penjejakan. Projek ini menyelesaikan cabaran ini dengan membina subsistem inventori. Sistem menggunakan metodologi Kitaran Hayat Pembangunan Sistem dan Rangka Kerja Senibina Kumpulan Terbuka (TOGAF) untuk analisis Seni Bina Perusahaan. Penyelesaian yang dicadangkan akan membantu UTHM beralih daripada persekitaran warisan yang tegar kepada seni bina Tiga Peringkat berasaskan web yang boleh berskala yang dihoskan dalam Pusat Data universiti. Sumbangan utama sistem termasuk pangkalan data berpusat masa nyata, antara muka responsif mudah alih untuk kebolehcapaian dan Modul Pembaikan khusus yang mengautomasikan laporan dan penyekatan aset yang rosak melalui pengimbasan kod QR. Deisgn akhir menyediakan pelan tindakan untuk sistem perusahaan bersatu yang meningkatkan pengalaman pengguna, memastikan pematuhan yang bersemangat dan menyokong UTHM dalam kecemerlangan operasi.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Universiti Tun Hussein Onn Malaysia (UTHM) is currently undergoing a project to build a new on-premise data centre to enable the consolidation, modernisation and management of its institutional IT infrastructure. Despite this strategic promise, the institution lacks formalised Enterprise Architecture (EA) capable of providing a unified structure that integrates business processes, information systems, data assets, and underlying technical platforms. The absence of formalised Enterprise Architecture has led to fragmented system implementations, data repositories, limited visibility of inter-system relationships, as well as significant challenges in realising interoperability and successful data integration among most departments and functional areas. These factors hamper the efficiency of operation, limit the use of data-driven decision making, and the ability of UTHM to achieve sustainable digital transformation. Compensating these shortcomings, UTHM has shown readiness to welcome external and academic inputs that can offer a comprehensive and versatile EA model that can be utilized in the institutions.

This thesis satisfies the institutional need by creating an overall Enterprise Architecture of UTHM based on The Open Group Architecture Framework (TOGAF). TOGAF is used because it has a universally recognized systematic approach, governance orientation, and suitability of large and complex organizations such as higher educational institutions. The suggested EA is to establish a standardised and unified architecture that supports the strategic objectives of UTHM in relation to its technological capabilities, maximises integration between the academic and administrative systems, and creates a better way of governing and managing the life cycles of the IT resources in addition to making future digital activity scalable. The

paper also develops a prototype enterprise system based on the use of the SAP technology to operationalise and test the proposed design. This prototype shows how key architectural components can be practically implemented, demonstrates how enterprise resource processes can be integrated into the EA framework, and evaluates the feasibility of SAP as the basic platform on which UTHM can base its mission-critical operations.

The proposed research aims at providing UTHM with a systematic, pragmatic, and future oriented Enterprise Architecture model through the incorporation of architectural design and system prototyping, which would facilitate the institutional governance, promote the system interoperability, enhance data integration and advance the long-term digital modernisation approach of the university in its new data centre setting.

1.2 Problem Statement

UTHM is currently planning to set up a new on-premises data center to facilitate the centralization and modernization of its institutional IT infrastructure. UTHM does not have an operational EA that reflects the manner in which its business processes, applications, data, and technology elements are consolidated and managed. This is due to the fact that UTHM does not have EA experience.

The absence of a clearly defined EA has led to a situation where there is a fractured system management, the inability to clearly see the system interdependencies, and the challenge of assuring the effective data integration. To address this issue, UTHM has indicated that it is receptive to outside suggestions, including academic donations, that may show an appropriate enterprise architecture model to be adopted.

To address this challenge, the case study will aim at enhancing the existing system landscape to design an effective and useful Enterprise Architecture of UTHM. Besides, a prototype system with the use of SAP technologies is created to enable UTHM to test an integrated enterprise system in the proposed architecture. This strategy seeks to show how the SAP can be aligned with the EA to facilitate the main operations of the university, enhance the integration of data, and confirm the feasibility of the suggested architecture in the new data center environment of UTHM.

1.3 Project Objectives

- To design a comprehensive architectural framework that aligns the inventory system with UTHM's organizational structure and ICT policies.
- To model and automate business processes that allow seamless handovers between departments.
- To establish a Single Source of Truth by designing an integrated enterprise data model and eliminating data redundancy and ensuring consistent asset information across all UTHM divisions.
- To develop a system capable of generating enterprise-level insights, helping UTHM management make data-driven decisions on budget planning and asset utilization.

1.4 Project Scope

The scope will fit within the project within which this project aims at offering a structured change within UTHM current enterprise model to an integrated enterprise model as per their objective in digitalizing transformation. The key elements of the project are:

1. Review existing system enterprise model

This project will start with an As-Is evaluation of the present system environment of UTHM according to the visit institutional observations. This step is aimed at examining the impact of redundancy of processes and inability to share real-time data on the efficiency of operations. The objective is to capture the existing relationship that exists between business processes and also analyze the effects of the existing system that is unintegrated on the efficiency of operations in the institutions.

2. Design To-Be Enterprise Model

The project entails the development of a full-blown "To-Be" Enterprise Model that will give a scaled and integrated plan of the university that charts the four layers of architecture (Business, Data, Application, and Technology) The scope of the project includes development of Use Case Diagrams, Use Case Descriptions, Activity Diagrams, and Sequence Diagrams to model the behavioral logic of the subsystems. Also it discusses Database Design using Class Diagram and Entity Relationship Diagram (ERD) in order to guarantee integrity of data and the interface design in order to determine the user experience and interaction of the individual subsystem in the new architecture.

3. System Development

The other part of this project is to develop a working prototype based on the SAP Business Technology Platform to make the various departments interconnected and demonstrate how the information flows freely. At this phase, the physical database is developed where all the required data is stored and the principal system functions are written. This process illustrates how an automated system that is integrated can ensure that university operations are faster and more efficient. Project Importance

1.5 Project Importance

The importance of this project is that it assists UTHM in solving the existing issues particularly in the enterprise architecture of UTHM concurrently to enhance its efficiency in terms of operations, system integration and user experience. The project guarantees the free flow of data among platforms such as Learning Management System (LMS) and student portal through analysis and improvement of subsystems integration such as student enrollment, course registration, hostel management and university healthcare. This is definitely positive as it will avail the students, staff and administrators with credible access to information and services.

Optimising proposal enterprise architecture helps UTHM make better decisions, use resources more effectively, and ensure their IT systems can grow over time. This approach will take full advantage of new technologies like AI, IoT, and cloud computing to support their academic, administrative, and research work. In the end, this project strengthens UTHM's commitment to digital transformation, operational excellence, and ongoing improvement, helping them remain a modern and efficient university.

CHAPTER 2

LITERATURE REVIEW

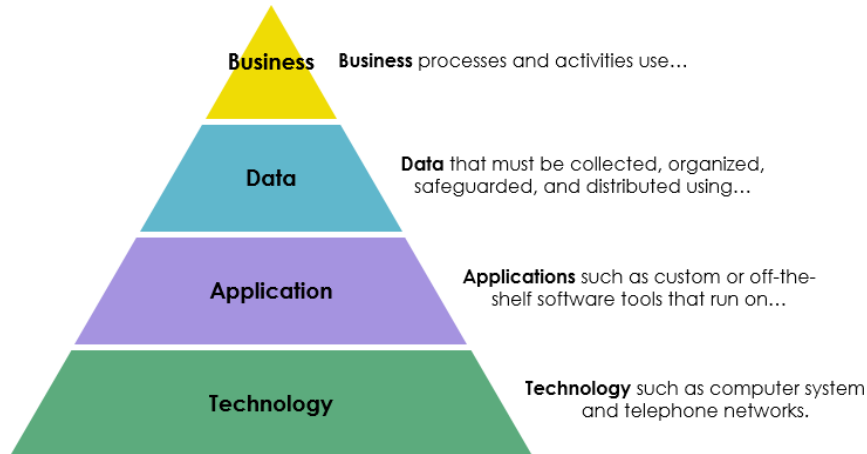
2.1 Introduction

This chapter provides a literature review of the key concepts that define the enterprise-level design and integration discussed in this project. The modern enterprise, especially in higher education, requires a comprehensive framework to strategically align its business objectives with its Information Technology (IT) capabilities and infrastructure. This framework is known as Enterprise Architecture (EA). We will also discuss the related sub-system that exist in universities EA by taking UiTM as an example. In the end, we are going to discuss technology used in EA which is the SAP Business Technology Platform and its details.

2.2 Enterprise Architecture (EA)

Enterprise Architecture (EA) is defined as a comprehensive framework designed to manage and align an organization's business objectives with its Information Technology (IT) capabilities and infrastructure (BOC Group, 2023; CIO Index, n.d.). It encompasses the systematic design, planning, and management of the organization's technology, processes, and systems to effectively support its strategic goals and objectives (BOC Group, 2023). EA provides a holistic view of the enterprise by organizing its structure into four interconnected architectural domains (Visual Paradigm, 2024)

Figure 1: The Four Domains of Enterprise Architecture



Note. Source: (Visual Paradigm, 2024).

Business Architecture: This domain maps the organization's strategy, governance, core business capabilities, value streams, and processes. Its central focus is on defining the "what" and "why" of the business operations, ensuring that technological investments are driven by strategic needs (CIO Index, n.d.).

Data Architecture: This domain establishes the principles and structures for how the organization's data assets are to be collected, stored, managed, and utilized. It is critical for ensuring data quality, consistency, and accessibility throughout the entire enterprise (BOC Group, 2023).

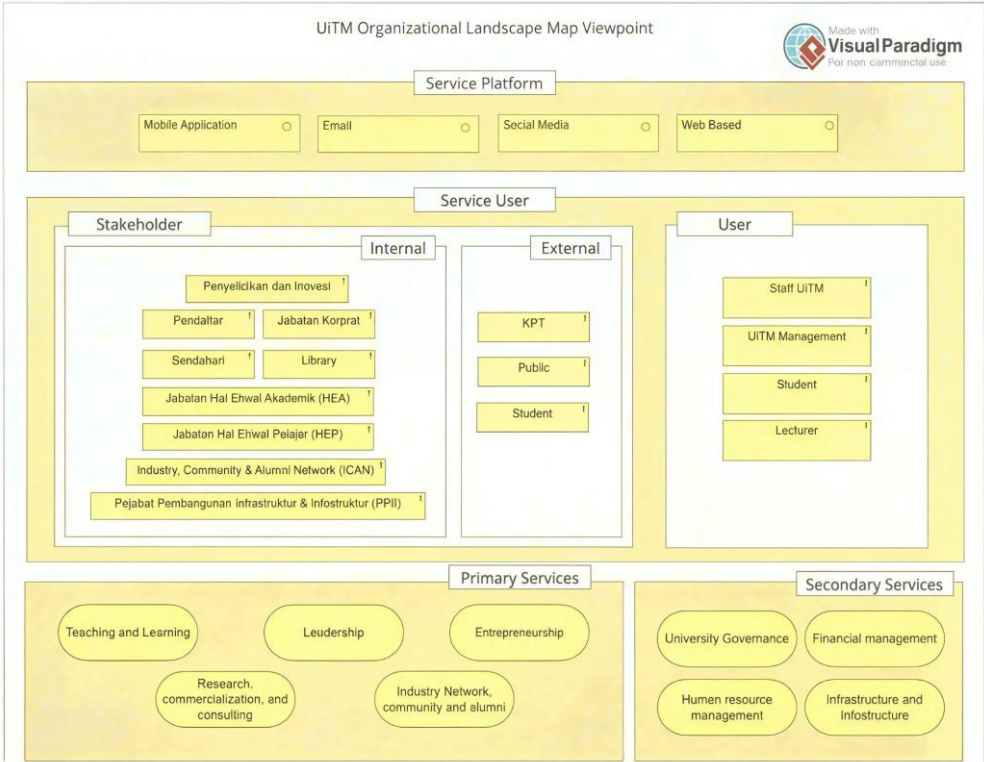
Application Architecture: This domain involves identifying the specific applications utilized within the enterprise and defining their interactions with each other and with external stakeholders. The objective is to establish an integrated, efficient, and scalable application portfolio (CIO Index, n.d.).

Technology Architecture: This domain describes the foundational hardware, software, networking infrastructure, and operating systems that collectively support the application and data layers. Its primary concern is ensuring the reliability, scalability, and security of the underlying technology foundation (BOC Group, 2023; CIO Index, n.d.).

2.3 Related Subsystem

Enterprise Architecture (EA) in higher education institutions typically integrates multiple subsystems that support academic, research, governance, administration, and community functions. These subsystems interact across different layers, including business processes, service delivery, information technology platforms, and stakeholder interactions, in an attempt to align institutions with their strategic goals. Many universities adopt EA for simplifying services provided to students and staff, standardising information management, and increasing decision-making effectiveness (Amin et al., 2024).

Figure 2: UiTM Organizational Landscape Map Viewpoint



Note: Source: (Amin et al., 2024).

Figure 2.2 shows UiTM’s Enterprise Architecture (EA) landscape, which is built around different subsystems. Each subsystem handles specific tasks, such as academic services, administrative services and digital platforms. They all work together to connect stakeholders and service users. The framework includes internal and external stakeholders, main and supporting service areas and works across platforms like mobile, email, social media, and websites. By linking these subsystems, UiTM creates

a smooth and integrated service environment that supports its goals for improving the institution (Amin et al., 2024).

Subsystem Category	Subsystem	Description
Service Platforms	Mobile App, Email, Social Media, Web-based Systems	Digital channels enabling communication and service access for users.
Stakeholders	Internal: HEA, HEP, Library, Corporate, ICAN, PPII, etc.	Organizational units involved in academic, student, infrastructure, alumni, and corporate services.
	External: KPT, Public, External Students	External bodies influencing or consuming UiTM services.
User Groups	UiTM Staff, Management, Students, Lecturers	Primary service users interacting with UiTM digital and administrative services.
Primary Services	Teaching & Learning, Leadership, Entrepreneurship, Research & Commercialization, Alumni & Industry Network	Core mission-driven services supporting UiTM's academic and developmental goals.
Secondary Services	University Governance, Human Resource, Finance, Infrastructure	Supporting subsystems enabling operations, compliance, and resource management.

Table 2.1 Description of UiTM EA Subsystems

Table 2.1 provides an overview of how each subsystem at UiTM supports the services offered at the university and the involvement of different stakeholders. It categorises subsystems into functional types, describes their purpose, and identifies service platforms on which users can access them. It also provides information on the internal and external stakeholders and user groups, and the primary and secondary services each subsystem enables. This structured view shows how UiTM integrates its digital and administrative systems into a seamless, mission-driven service environment.

Feature	Layered EA Model	UiTM Organizational Landscape Map Viewpoint
Main Focus	Architectural Components (What are the components of the enterprise's IT?).	Organizational and Service Components (Who uses the services and what services are offered?).
Business/Organizational Content	Lists Functional Areas : Academic Management, Finance, Human Resources, etc.	Lists Key Stakeholders (Internal/External), Users (Staff, Student, Lecturer), and Services (Teaching, Leadership, Governance).
IT/Technology Content	Lists Abstract Domains : Servers, Network, Middleware, Database (under Technology Architecture).	Lists Delivery Channels/Platforms : Mobile Application, Email, Social Media, Web Based (under Service Platform).
Framework Link	Strongly aligned with the TOGAF/Zachman frameworks' core layered structure.	Represents a Viewpoint or business-centric model, often part of an overall EA deliverable.

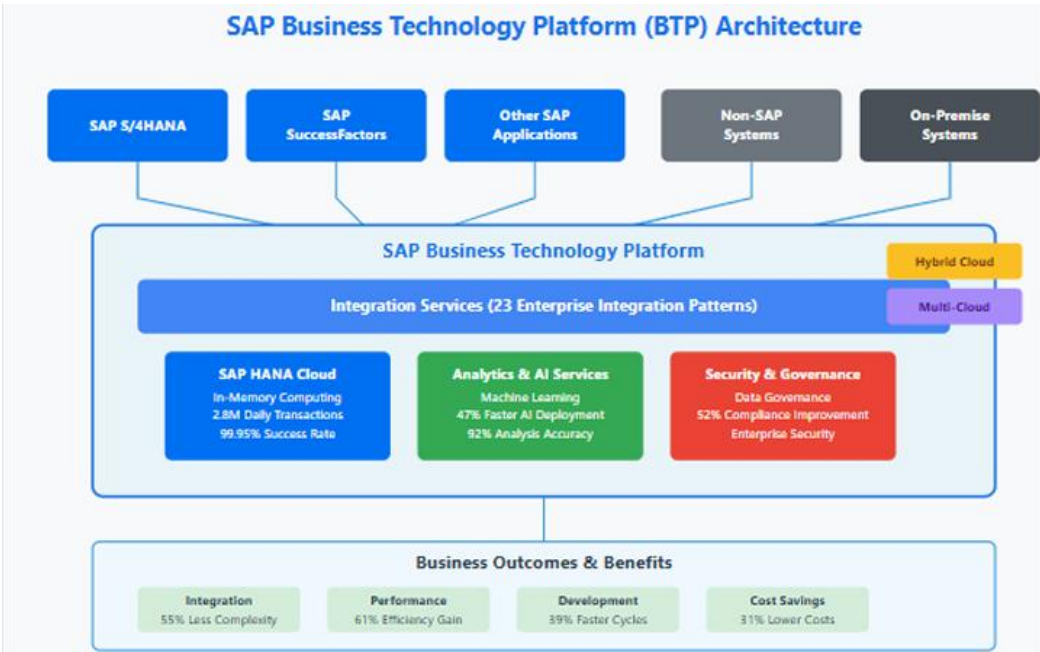
Table 2.2 Comparison between Layered EA Model and UiTM Organizational Landscape Map Viewpoint

This Table 2.2 visual two representations of Enterprise Architecture (EA) serve distinct but complementary roles: the Layered EA Model acts as a structural blueprint, primarily focusing on Architectural Components by detailing the required Functional Areas like Academic Management and the underlying abstract IT Domains such as Servers and Network, consistent with the formal, tiered structure of frameworks like TOGAF; conversely, the UiTM Organizational Landscape Map Viewpoint functions as a high-level, business-centric communication tool, focusing on Organizational and Service Components by illustrating the key Stakeholders (Staff, Students) and the Delivery Channels (Mobile Apps, Web) they use to access the university's services.

2.4 Technology Used

The SAP Business Technology Platform (BTP) is architecturally designed as a unified cloud foundation, acting as a crucial integration layer for the modern enterprise (Komarina & Sajja, 2025). It is built upon a framework that encompasses the Integration Suite, Extension Suite, and Data Management Suite, leveraging a microservices approach to provide a platform with defined scalability features and a structured security framework (Annanki, 2023).

Figure 4 : SAP BTP Architecture



Note: Source: (Komarina & Saja, 2025).

Core Components of SAP BTP

1. Database and Data Management

Database and data management focuses on efficient data storage, processing, and governance. Key services include SAP HANA Cloud, SAP Data Warehouse Cloud, and SAP Data Intelligence, which collectively support real-time data access and enterprise-level data management.

2. Analytics

SAP BTP includes analytical tools such as SAP Analytics Cloud (SAC), enabling organizations to perform reporting, data visualization, predictive analytics, and planning. These tools help businesses make informed, data-driven decisions.

3. Application Development and Automation

This component offers capabilities for building and extending applications using low-code/no-code tools, cloud-native development frameworks, and process automation. Tools such as SAP Build Apps, SAP Build Process Automation, and the SAP Cloud Application Programming Model (CAP) streamline application development and workflow automation.

4. Integration

SAP BTP supports seamless connectivity between SAP and non-SAP systems through services like SAP Integration Suite, API Management, and Event Mesh. These tools ensure smooth data exchange and interoperability across the organization's technology landscape.

SAP BTP enables organizations to modernize business processes, reduce dependency on custom code in core systems, and create flexible, scalable solutions. By unifying application development, data management, analytics, and integration services, the platform helps businesses enhance innovation, improve efficiency, and adapt quickly to changing operational needs.

2.5 Summary

This chapter establishes the theoretical foundation for the project by exploring Enterprise Architecture (EA) as a tool to align university business goals with IT infrastructure. By examining the four key domains which are Business, Data, Application, and Technology, the review clarifies how a system must be structured to support an entire organization.

Through the case study of UiTM's EA landscape, the chapter illustrates how various subsystems and stakeholders interact to create an integrated service environment. Finally, the review of SAP Business Technology Platform (BTP) identifies the modern technology needed to achieve this integration, focusing on data management, analytics, and seamless system connectivity. Together, these elements provide the blueprint for designing the UTHM Inventory System as a robust enterprise solution.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter is a description of the methodology of Waterfall with System Development Life Cycle (SDLC) used to help with the planning, analysis, design and implementation of improvements to UTHM's Enterprise Architecture framework. The Waterfall of SDLC uses formal, creating or altering systems, and the models and methodologies that people use to develop these systems (Kute & Thorat, 2014). The choice to use a structured methodology focused on requirements, architectures and specifications being fully defined and validated earlier in the process was due to the increased difficulty in changing or correcting Enterprise Architecture projects after they have been implemented due to their high cost of implementation and complexity to implement from start to finish. The following chapters explain each phase of the methodology, including phases, activities and deliverables, provide a project schedule like for example using a Gantt Chart or Kanban format with major milestones, and summarise the entire development of the project.

3.2 Proposed Methodology

This study used the Waterfall System Development Life Cycle (SDLC) model as its approach. The Waterfall methodology is chosen for its systematic, documentation-centric, and linear characteristics, making it appropriate for major business projects like the creation of business Architecture (EA) for Universiti Tun Hussein Onn Malaysia (UTHM). Considering that EA projects entail substantial implementation costs, widespread institutional effects, and long-term strategic coherence, a methodology prioritising meticulous planning, requirement verification, architectural delineation, and regulated execution is necessary. The Waterfall SDLC facilitates thorough analysis and robust architectural design prior to deployment, reducing rework and maintaining architectural consistency throughout the project lifecycle.

This study uses the Waterfall technique to direct the design and development of an improved Enterprise Architecture framework for UTHM, alongside the creation of an enterprise-wide SAP prototype to validate the proposed architecture. Each phase of the technique concentrates on systematically converting UTHM's institutional requirements into a structured design, subsequently validated through technical implementation. Stakeholder interaction is integrated to guarantee conformity with institutional expectations and operational reality. The suggested methodology has five primary phases: Feasibility Study, System Analysis, System Design, Implementation, and Maintenance and Support.

3.3 Phases of Methodology

The project follows the five core phases of the SDLC which is Feasibility Study, System Analysis, System Design, Implementation and Maintenance and support

Phase 1: Feasibility Study

This first stage will determine whether the project is viable. We adhere to UTHM's standard operating procedure for new system requests.

- FSR Process: The project began with the submission of the Feasibility Study Report (FSR).
- Committee Approval: FSR is submitted to the IT Security & Steering Committee which will evaluate the proposal.

Outcome: The "Go /No-Go" decision based on the project proposal and literature review.

Phase 2: System Analysis

In this phase, we collect the detailed business requirements to bridge the gap between the definition of the requirement and how it should be.

- Data Collection: Conducting interviews with UTHM stakeholders (Asset Unit officers) to understand the current manual form and workflow.
- Requirement Specification: Defining the user roles and their access privileges.

Phase 3: System Design

In this phase , we convert the requirements into technical specifications.

- **Architecture Design:** We utilize the three-tier architecture model to ensure the system is scalable and easy to maintain.
- **Database Design:** Create the entity-relationship diagram to model the relationships between asset, staff and location.
- **Interface Design:** Prototyping the dashboard UI to ensure they are user-friendly for non technical staff.
- **Infrastructure Design:** The system will be hosted in UTHM's Data Center. It will benefit from high-availability infrastructure that includes dual power sources, UPS backup and generator set to ensure very high uptime.

Phase 4: Implementation

This is the phase where the system is built and implemented.

- **Coding:** Developing the subsystem using web technologies such as PHP and java then connect it to the database MySQL.
- **Data Migration:** Adopt a phased migration strategy, which converts sample data from existing Excel sheets into the new system format.
- **Security Implementation:** The code will be hosted behind the UTHM's existing security layer which include 2 Firewalls (Europe/China) and a Web Application Firewall to protect the portal from being attacked from the external side.
- **Testing:** Perform testing on each individual module and integration testing to ensure it communicates with the Financial module.

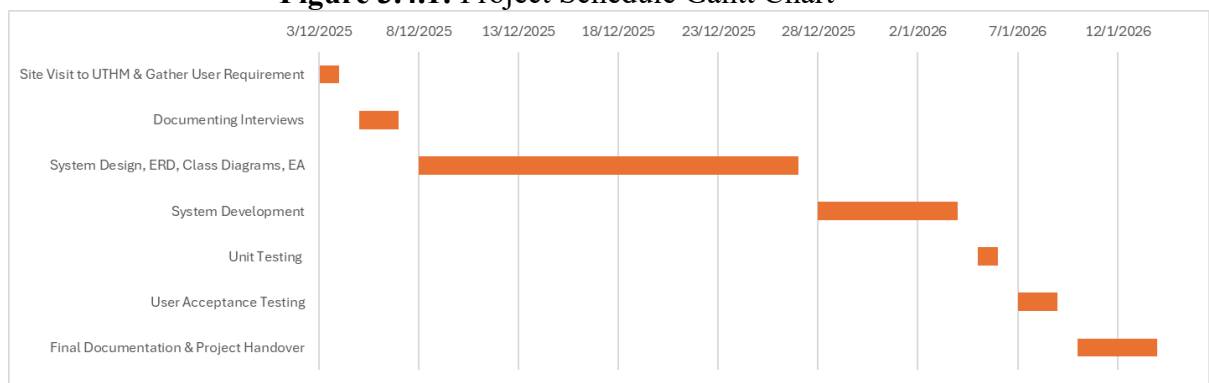
Phase 5: Maintenance and Support

In this phase , we involve training and patching activities. We will teach the UTHM staff how to use the new QR code features and fix any bugs that are discovered after the system is deployed.

3.4 Project Planning Schedule

In order to make sure this project runs smoothly, a schedule consists of every phase and its duration is made. The team members that are in charge of leading the task are also determined to make sure all the phases are supervised.

Figure 3.4.1: Project Schedule Gantt Chart



Tasks	Start Date	End Date	Task Leader
Site Visit to UTHM & Gather User Requirement	3/12/2025	4/12/2025	Aisyah
Documenting Interviews	5/12/2025	7/12/2025	Harini
System Design, ERD, Class Diagrams, EA	8/12/2025	27/12/2025	Zhiming
System Development	28/12/2025	4/1/2026	Afiq
Unit Testing	5/1/2026	6/1/2026	Auni
User Acceptance Testing	7/1/2026	9/1/2026	Ziyaad
Final Documentation & Project Handover	10/1/2026	14/1/2026	Ika

3.5 Summary

This chapter has elaborated the Waterfall System Development Life Cycle (SDLC) methodology that was selected to inform the enhancement of the Enterprise Architecture at UTHM. Through the five essential stages that include feasibility study, system analysis, system design, implementation and maintenance, the project reduces the risks of the high cost and complexity of enterprise level systems.

This organized approach will guarantee that the final SAP prototype is not only a sound technical prototype but also a strategic one that is capable of supporting the operational needs of UTHM in its new data center setting and is safe. Inclusion of a detailed project schedule and Gantt chart will help to ensure that all activities such as visiting the site to the implementation and testing of the system all fall within the stipulated time. Finally, this systematic approach is the key to the transition between the strategic digital transformation objectives at UTHM and the achievement of a secure, integrated, and high-performance enterprise environment.

CHAPTER 4

Analysis and Design

4.1 Introduction

This chapter examines the organisational structure of Universiti Tun Hussein Onn Malaysia (UTHM), the present system environment, and business needs to support the creation of a full Enterprise Architecture (EA) framework and a SAP-based enterprise prototype. The overall understanding of the organisational framework, the work process, and the methods that the university uses to disseminate information is paramount in making sure that the proposed design is aligned with the objectives of the institution and that it addresses the current problems in the operations of the organisation. Since UTHM currently operates with a number of independent systems with very little integrative aspects, the analysis forms the necessary basis of defining how an integrated and centralised enterprise structure can enhance the efficiency of the institution, its governance, and its use of data to make informed decisions.

The chapter first examines the organisational structure of UTHM at Strategic, Tactical and Operational levels in order to identify key decision-makers, stakeholders and key data-creators. This is followed by a thorough analysis of the current system environment, with a focus on the reliance on fragmented digital environments and semi-manual processes, despite having a modern three-level data centre environment. This is followed by a comparative analysis of the present environment and the proposed environment so as to illustrate the transition between isolated systems to a fully integrated environment of the enterprise with the help of centralised data management, Single Sign-On (SSO) and a better user experience.

In order to ensure that the proposed architecture is reflective of institutional requirements, the requirements gathering activities were conducted in terms of stakeholder participation by way of structured interviews. Through these interactions, the identification of the critical business needs, business challenges, technical expectations, and governance considerations became possible. The chapter concludes by outlining the functional and non-functional requirements that will guide the design, development and implementation of the enhanced EA framework. These studies all form the analytical basis of the subsequent design and solution implementation stages of architecture.

4.2 Company Organization Structure

A successful system integration requires clear understanding of the organizational hierarchy and the information flow between stakeholders. UTHM operates on three distinct levels which are Strategic, Tactical and Operational. This structure must be bridged to eliminate the "Vertical Silos" which are often found in large organizations.

1. Strategic Level

- IT Security and Steering Committee: This high-level body responsible for the overall digital strategy for UTHM. They will review and approve the Feasibility Study Reports for all proposed system. They need to determine whether the new system, is necessary and aligns with the university budget and culture.
- Management Implications: Top management support is critical to enforce the cultural changes to adapt the new system.

2. Tactical Level

- Bursary Department (Bendahari) : Bursary defines the rules that the software must obey. They act like a central hub for data verification, requiring inputs from various faculty to ensure accuracy.

3. Operational Level

- Faculties and Responsibility Centers: This level includes academic and administrative staff that are distributed across the campus. They are the primary data generators.

4.3 Current System Analysis

UTHM is a university that uses various digital systems and technologies to facilitate its academic, administrative, and operational processes. Some of the aspects taught and learned through online platforms include student administration, finance, human resource management, and communication between students and staff. Such systems allow prosaic access to information and routine. They are however implemented on their own in order to satisfy departmental requirements. This has led to the restricted data sharing between departments and reliance on manual operations to increase operational efficiency. With regard to infrastructure, UTHM is still developing a new three-tier data center that will accommodate and support its digital systems. Though such infrastructure is a powerful physical backbone of digital services, the capabilities of such infrastructure are not fully applied in the present system environment because there are no common system management and integrated digital solutions. This case indicates the necessity to improve the coordination of the system and the orderly planning of the maximization of the advantages of the new data center.

4.4 Comparison between Existing and Proposed System

The change of the current digital environment of UTHM to the proposed architecture constitutes a change in a disjointed system to an integrated digital system. The university is at the moment dependent on autonomous systems which cater to departmental requirements. Although there is a contemporary three-tier data center, its potential is not fully exploited as the existing systems are not centralized and integrated to exploit such a base. The suggested system will overcome these constraints by substituting manual workarounds with full-scale digitalization and a clear vision of the Digital Transformation. The proposed architecture will present a centralized data warehouse and automated pipelines instead of isolated data as a way of ensuring information flows smoothly throughout the institution. This shift is to a more nimble, hybrid cloud setup. Also, the user experience is enhanced when using many different disconnected services to a simplified solution that has Single Sign-On (SSO) and a single mobile application.

System Requirements Gathering Technique

To ensure the proposed UTHM Inventory System aligns with the actual operational needs of the university, the team employed the Interview Method as the primary data collection technique. This qualitative approach allowed the team to explore the complexities of existing workflows and identify specific pain points across different hierarchical levels. The interview method was selected due to its flexibility and the depth of information it provides. Compared to other methods like questionnaires, interviews allow the team to capture the specific needs of both the administrative staff and the end-users. The interviews were conducted during the site visit on 3/12/2025.

The team roles were distributed as follows:

- **Lead Interviewer (Aisyah):** Facilitated the discussion using a prepared set of open-ended questions.
- **Technical Observers (Afiq & Ziyaad):** Focused on identifying technical constraints and integration points with current UTHM legacy systems.
- **Note-Takers (Ika & Harini):** Documented the responses and captured non-verbal cues regarding system frustrations.

Below are the sample of questions interviewed and its answers:

Q: Are there still manual methods or all is digitalized?

A: UTHM is not 100% digital yet. One of the departments that are still running manually are their Health Center. However, admin been enforcing lecturer to use LMS and work on a singular system.

Q: Is there a stand alone database different from the main database?

A: Only LMS has separated database but all the system get data in on single database

Q: Who approves of the new system?

A: The team asks for user requirements first, then the team will develop the system and send it to the committee to decide either to accept or reject the new system.

4.5 System Requirements

Functional Requirements

Functional requirements specify the particular functionality and features that an updated enterprise architecture (EA) framework should offer to UTHM to help streamline primary business operations in UTHM's IT ecosystem. Functional requirements help to ensure that the updated EA framework provides support for key UTHM processes such as asset management, decision making and compliance.

- User Authentication - A secure account accessible for all UTHM staff and students using Single Sign-On (SSO).
- Requirements Management - All systems requirements should be captured, traced and validated from the SDLC analysis phase and then linked to the design output.
- Reporting - Generate customizable reports that should be exportable along with Gantt charts and phase deliverables that can be viewed by administrators.
- Integration Support - EA Framework must provide the ability to import and export data that is in existing tools like TOGAF during integration activities.

Non-Functional Requirements

Non-functional requirements specify the quality attributes and performance standards the EA framework must meet, ensuring reliability, efficiency, and usability in UTHM's demanding academic environment.

- Performance: System shall load web pages within 3 seconds for datasets up to 1,000 components.
- Scalability: System must support around 2000 concurrent users.
- Security: Encrypt data especially for password and using firewall like Web Application Firewall (WAF) on the webpage system.
- Usability: Intuitive for new users and accessible for various users.
- Reliability: The system must have 99.5% uptime.

4.6 System Design

4.6.1 Enterprise Architecture AS-IS

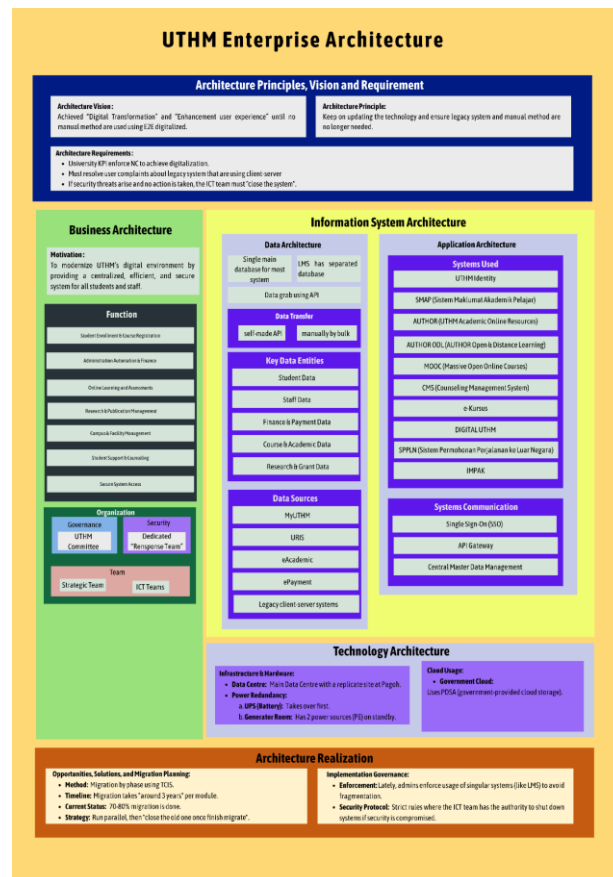


Figure 4.1 UTHM Enterprise Architecture AS-IS

4.6.2 System Design AS-IS

This section explains the current architecture at UTHM before the new system is implemented.

1. Business Architecture

- **Strategy:** Guided by "Digital Transformation 2030" to shift the culture from manual to digital, but administrative tasks still require manual forms currently.

- **Governance:** All proposed systems must pass the IT Steering Committee via a Feasibility Study Report and approved based on necessity and department expertise.
- **Operations:** Departments operate in horizontal silos. The LMS and TCIS are currently run as separate business entities with disconnected workflow.

2. Application Architecture

- **Core system:** Legacy TCIS based on client-server technology.
- **Accessibility:** Restricted on campus LAN. Staff cannot access the system while outstation without a VPN , this leads to bad user experience.
- **Integration:** Applications are fragmented. For example the LMS is a separate application from the main student database, requiring API to bridge the gap.

3. Data Architecture

- **Storage:** Data is distributed and decentralized. For example the student marks and physical asset record are separate, siloed databases.
- **Synchronization:** The data transfer relies on API calls or need manual entry , rather than real-time synchronization.
- **Performance:** Current database structure struggles with high concurrency or usage, causing up to 5- minute lags during peak usage.

4. Technology Architecture

- **Facility:** Tier 3 Data Center located on the upper floor (flood prevention).
- **Server Hosting:** Uses Hot Aisle/Cold Aisle Containment for thermal efficiency.
- **Power:** High availability via Dual Power Sources (TN A/TN B), backed by UPS and Generator Sets.
- **Security:** Perimeter protection using 2 Firewalls (Europe/China brands) and a Web Application Firewall (WAF) for portal defense.

4.6.3 Enterprise Architecture To-Be

Figure 4.2 illustrates the Enterprise Architecture To-Be for UTHM. It represents the future system structure of the university. This architecture helps improve system efficiency and supports UTHM's future digital development.

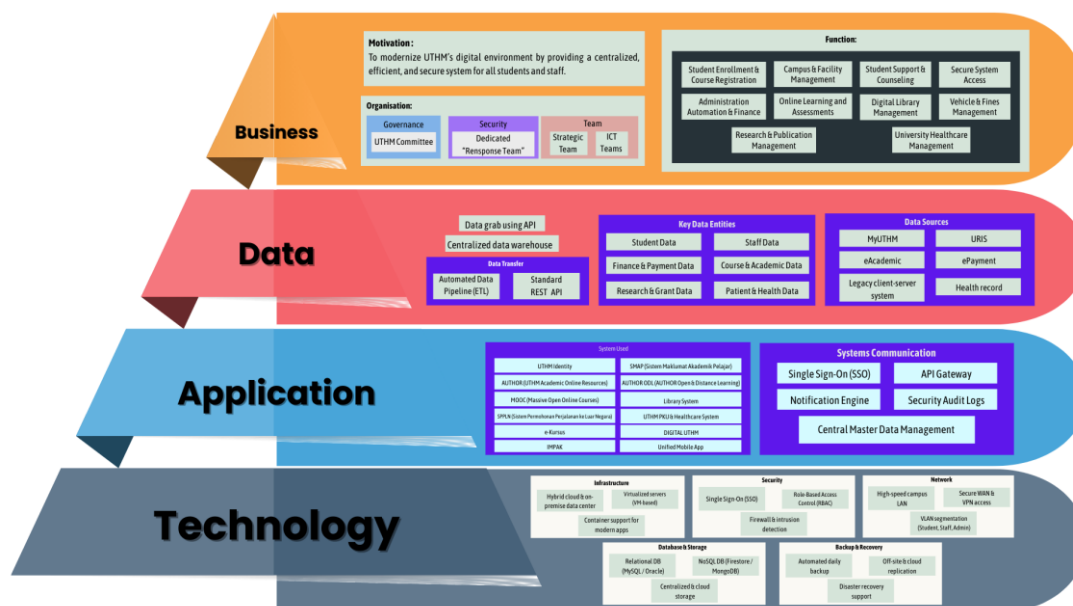


Figure 4.2 UTHM Enterprise Architecture To-Be

4.6.4 To-Be

This section explains the Enterprise Architecture To-Be as shown in Figure 4.2. The framework is organized into several main layers using TOGAF (The Open Group Architecture Framework). Each layer serves a specific purpose.

1. Business Architecture

This layer, shown in green, describes what the university does and how it operates:

- **Motivation:** To develop a centralized, secure and efficient digital environment accessible to students and staff on any device at any time.
- **Organization:** Illustrates UTHM's departmental hierarchy.
- **Functions:** The architecture supports ten core business functions:

Business Functions	Description
Student Enrollment & Course Registration	Manages student admissions, course selection, and registration processes in a streamlined manner.
Administration Automation and Finances	Manages administrative tasks, payroll, payment processing and allocation of university resources.
Online Learning and Assesment	Support student learning and coursework using online platform.
Research and Publication Management	Supports research project management, grant applications and innovation initiatives.
Campus and Facility Management	Coordinates maintenance, space allocation, and utilization of campus facilities.
Student Support and Counselling	Manages student welfare services, counselling sessions, and academic support programs.
Secure System Access	Authorized access to systems through identity management and authentication mechanisms.
University Healthcare Management	Supports the management of healthcare services and medical records within the university.
Digital Library Management	Provides access to digital academic resources, journals and library services.
Vehicle and Fines Management	Manages vehicle registration, parking systems and fine enforcement processes.

Table 4.1 Business Functions Description

2. Information System Architecture

Divided into Data Architecture and Application Architecture:

- **Data Architecture:** Includes a centralized data warehouse that consolidates data from multiple systems. It supports dashboards for data visualization, automated data pipeline and data protection mechanisms. REST APIs are used to enable secure and standardized data exchange between systems.
- **Application Architecture:** The Application Architecture consists of multiple application layers that support academic, administrative, and operational activities at UTHM.

1. **Student & Academic Systems:** UTHM Identity, SMPA, MyUTHM Academic, Open and Distance Learning (ODL) platform, Massive

Open Online Courses (MOOCs), Author ID, and online academic resources.

2. Administrative & Support Systems: Counselling Management System (CNS), SPLPN, UTHM Healthcare System, e-Kursus, DIGITAL UTHM, IMPAK, and Vehicle and Facilities Management systems.
3. Library & Access Systems: Library System and Unified Mobile Application, providing centralized access to academic and student services.
4. Integration: Connects all systems through Single Sign-On (SSO), API Gateway, notification engine, security audit logs, and centralized master data management

3. Technology Architecture

Defines the infrastructure supporting all systems and is organized into seven domains:

- Infrastructure: Hybrid cloud access and on-premise data center with virtualization support to ensure scalability, flexibility and efficient resource utilization.
- Network: High-speed backbone network, secure Wan and VPN access to support secure and advanced connectivity across the campus.
- Platform: Linux and Windows servers and web and mobile applications to support the deployment and execution of modern applications..
- Database: Relational (MySQL/Oracle), NoSQL (Firestore/MongoDB) and cloud storage for high availability.
- Application Hosting: Web servers such as Apache and Nginx, cloud-based hosting services, and API Gateway to host and integrate university applications.
- Security: Single Sign-On (SSO) , secure internal communications, firewalls, and intrusion detection to protect applications, data, and network resources.
- Backup: Automated backups, offsite/cloud replication, and disaster-recovery support data protection and business continuity.

4.6.5 System Architecture

Figure 4.3 below illustrates the system architecture of the UTHM Inventory Subsystem. Users access the system through the Web Interface, and all requests are transmitted via the Internet. Users are first authenticated using UTHM Single Sign-On (SSO) before

any system functionality can be accessed. Once authenticated, requests are processed by the API Layer, which routes them to the appropriate application modules. These modules execute the required business logic and interact with the Inventory Database to retrieve or update data. The processed results are then returned through the API Layer to the frontend for display.

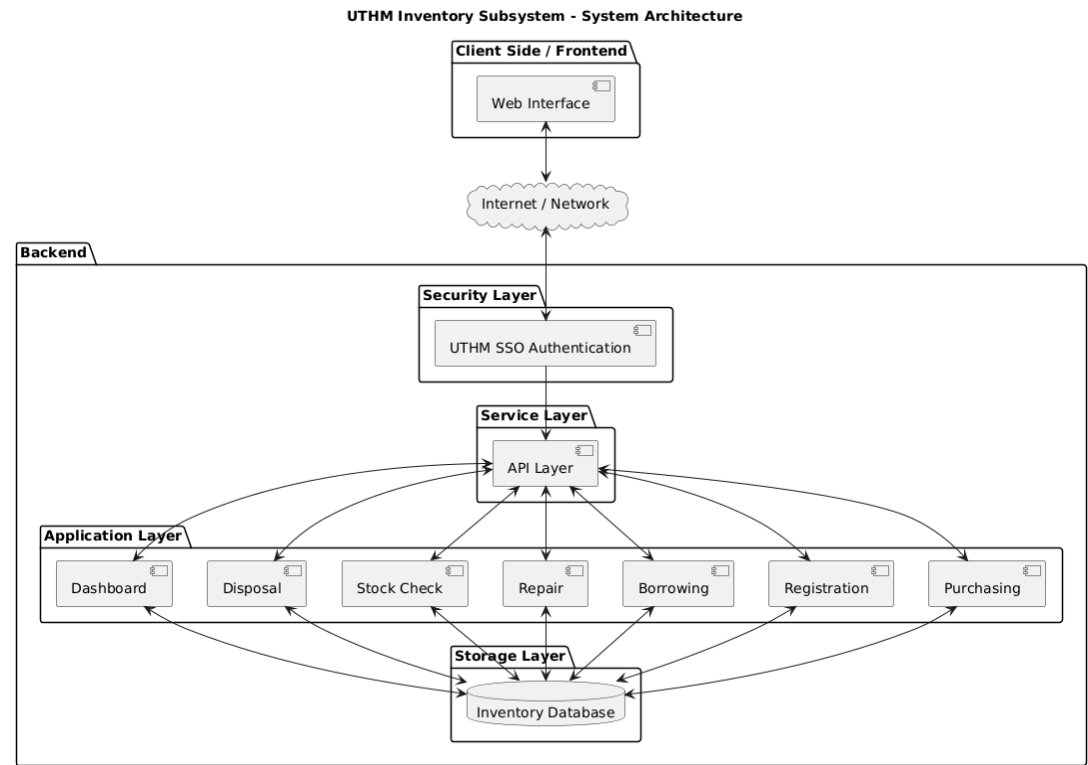


Figure 4.3 UTHM Inventory System Architecture

4.6.6 System Architecture Components

The UTHM Inventory Subsystem is designed using a cloud architecture. As shown in Figure 4.3, it divides the system into presentation which is the frontend, security, service, application and data layers. This design has a clear separation of responsibilities, maintainability and secure access to inventory data. The following explains the components in each layer and their functions.

1. Presentation Layer (Client Side / Frontend)

- Component: Web Interface
- Function: The presentation layer is the user-facing part of the system, where staff and students interact with the inventory subsystem. It handles input forms, dashboards and reports. It also presents all the data and insights in an intuitive and readable format. The frontend communicates with the backend over the network but does not perform any business logic or data storage operations.

2. Transport Layer (Internet / Network)

- Component: Internet or Network Cloud
- Function: This layer provides the connectivity between the frontend and backend. It enables secure transmission of requests and responses and allows users to access the system remotely while protecting data and confidentiality during transit.

3. Security Layer

- Component: UTHM SSO Authentication
- Function: Security is the first point of access for any system request. The UTHM Single Sign-On (SSO) verifies user credentials and enforces access control to ensure that only authorised personnel can perform operations such as purchasing, borrowing or stock checking. Authentication occurs before any business logic is executed to provide a secure foundation for the system.

4. Service Layer (API Layer)

- Component: API Layer

- Function: The service layer acts as a gateway between the frontend and the application modules. It receives authenticated requests from the user interface, validates requests and routes them to the appropriate business logic modules and returns processed results to the frontend. This separation ensures that the frontend does not directly access backend modules or databases. This way, security risks are reduced and modularity is achieved.

5. Application Layer (Business Logic Layer)

- Components: Purchasing, Registration, Borrowing, Repair, Stock Check, Disposal, Dashboard
- Function: The application layer contains the core business logic of the inventory system. It consists of eight functions:
 1. Purchasing: To adds new inventory items.
 2. Registration: To manages registration of users or items.
 3. Borrowing: To tracks check-outs and returns.
 4. Repair: To handles repair requests for damaged items.
 5. Stock Check: To provides real-time inventory status and reports.
 6. Disposal: To manages retirement of obsolete items.
 7. Dashboard: To summarizes key metrics for management.

All modules interact with the data layer to read or update inventory records that are being made by the administrator, lecturers or students of UTHM.

6. Data Layer (Storage Layer)

- Component: Inventory Database
- Function: The data layer stores all persistent system data including inventory records, user information and transaction history. It establishes data consistency, integrity and availability for all application modules.

4.7.8 Project Design

To model the functional requirement of Repair Subsystem, we utilize UML diagrams. These diagrams visualize how the system handle the lifecycle of broken asset in UTHM and moving it from “Avaliable” to “Under Repair” and finally back to stock.

1. Use Case Diagram for Repair Subsystem

Figure 4.7.7.1 illustrates the interaction between the primary actors and the Repair Subsystem.

The system involves two key actors which is the General Staff (lecturers or lab assistants) who initiate the defect reporting process, and the Technician, who is responsible for managing the maintenance workflow.

Use Cases:

Report Asset Defect: This use case allows the General Staff to submit a damage report. It includes a mandatory "Verify Asset ID" process to ensure the asset exists in the database before a report can be filed.

View Repair Queue: Enables Technicians to see a centralized list of all pending maintenance tasks.

Update Repair Status: Allows Technicians to modify the lifecycle of a ticket (e.g., from Pending to Fixed or Unrepairable).

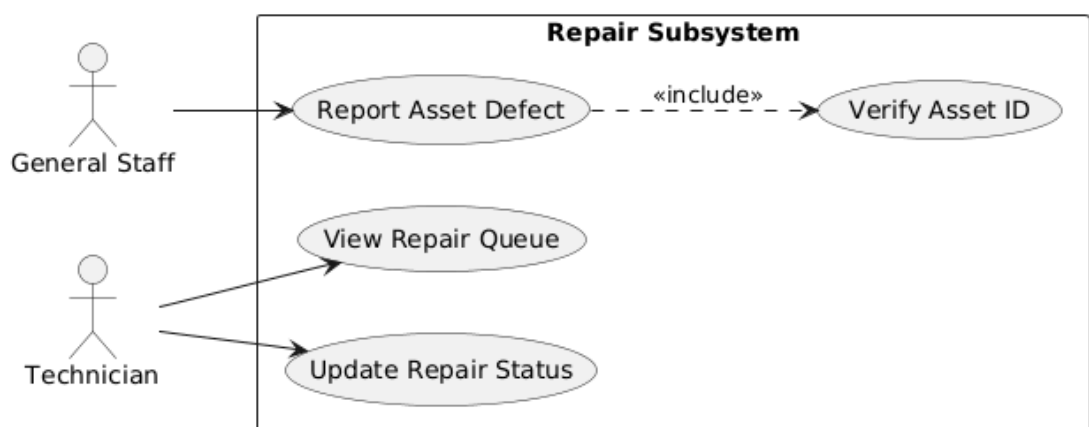


Figure 4.7.7.1 Repair Subsystem Use case diagram

2. Use Case Description

Field	Description
Use Case ID	UC REP 01
Use Case Name	Report Asset Defect
Actor	General Staff (Lecturer / Lab Assistant)
Description	Allows a staff member to report a broken asset so it can be blocked from borrowing and scheduled for repair.
Pre-Condition	1. User is logged in. 2. Asset status is currently "Available".
Flow of Events	1. User scans the Asset QR Code or searches the Asset ID. 2. System displays Asset Details. 3. User clicks the "Report Damage" button. 4. System prompts for "Damage Description" and optional photo upload. 5. User submits the form. 6. System updates Asset Status to "Under Repair" (Blocking borrowing). 7. System generates a Repair Ticket ID.
Post-Condition	Asset status is "Under Repair" and a notification is sent to the Technician.
Exceptions	If the Asset is already marked "Lost" or "Disposed", the system blocks the report.

3. Activity Diagram

Figure 4.7.7.2 show the end-to-end workflow of the repair process, highlighting the automated logic of the system.

- **Process Flow:** The activity begins when a staff member scans an asset's QR code. If the asset is identified as broken, they submit a "Damage Report."
- **System Automation:** A critical feature shown in the "System" partition is the Auto-Block Mechanism. Once a report is submitted, the system automatically changes the asset's status to "Under Repair" and blocks it from being borrowed, ensuring data integrity without human intervention.
- **Decision Node:** The workflow concludes with a decision point for the Technician. If the repair is successful, the asset is returned to the "Available" pool. If the repair fails, the asset is flagged as "Beyond Economic Repair," triggering the official disposal process (KEW.PA-19).

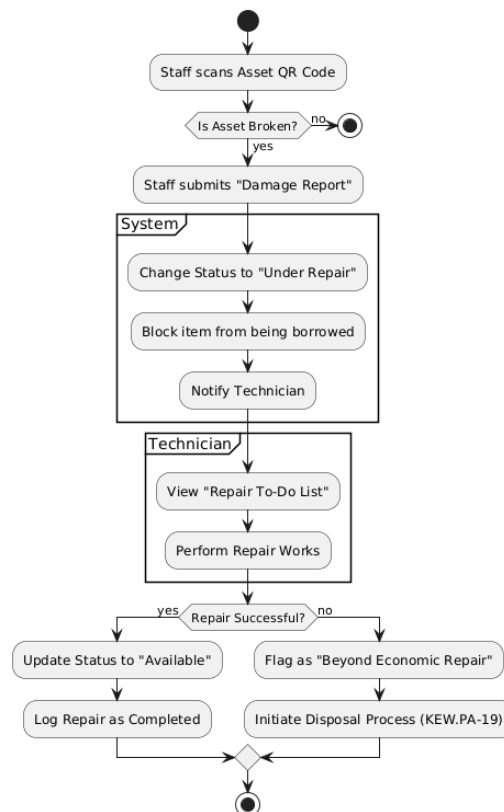


Figure 4.7.7.2 Repair Subsystem Activity Diagram

4. Sequence Diagram

Figure 4.7.7.3 visualizes the technical sequence of messages exchanged during the "Report Defect" scenario.

Object Interaction: The process starts when the User interacts with the RepairInterface by clicking "Report Damage." The interface forwards this data to the RepairController.

Database Operations: The controller performs three distinct database operations in sequence:

1. Validate: Checks if the AssetID exists.
2. Create: Generates a new RepairTicket linked to the user.
3. Update: Crucially, the controller sends an updateStatus command to the AssetDatabase, changing the asset's status to "Under Repair" immediately.

Feedback: Finally, a success message is returned through the controller to the interface, confirming to the user that the report has been lodged.

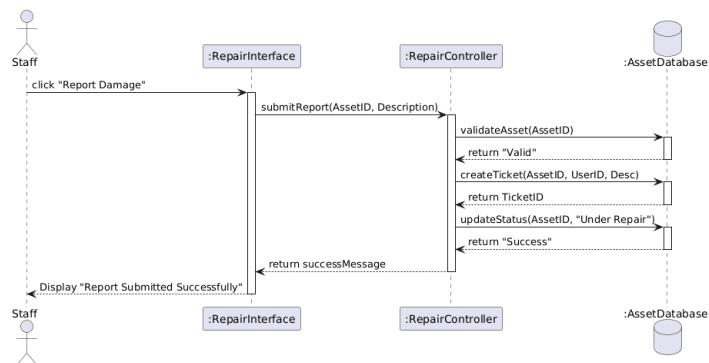


Figure 4.7.7.3 Repair Subsystem Sequence Diagram

4.7.8 Database Design

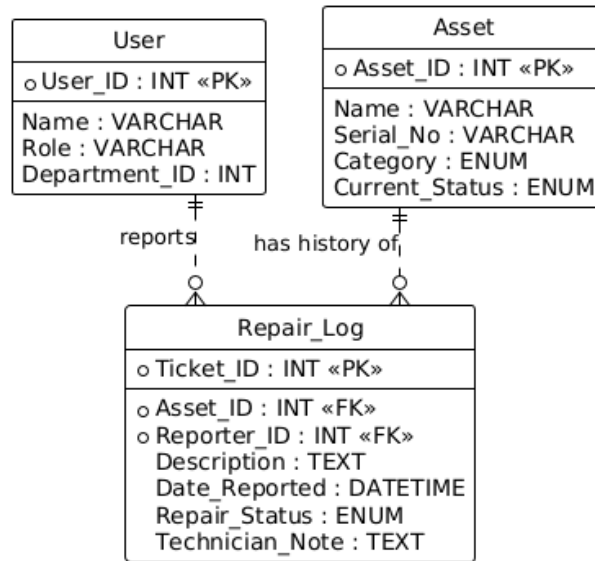


Figure 4.7.8.1 Repair Subsystem ERD

To support the repair module, we introduce a specific table to track the maintenance logs. This follows the Relational Database to ensure data integrity with our main inventory system.

Table Structure:

Table: Repair_Log

Ticket_ID (Primary Key, INT)

Asset_ID (Foreign Key linked to Asset Table)

Reporter_ID (Foreign Key linked to User Table)

Description (Text)

Date_Reported (DateTime)

Status (Enum: 'Pending', 'In-Progress', 'Completed', 'Unrepairable')

Technician_Note (Text)

Relationship:

One Asset can have Many Repair Logs (History).

One User can submit Many Repair Logs.

4.7.9 Interface Design

This figure is a main landing page of the UTHM Central Inventory System. It is the central navigation hub and makes a clear separation between the three core modules: the Staff Portal (to be used by general users to report issues), the Technician Dashboard (to be used by IT staff to manage repairs) and the Asset Stock view (to monitor total inventory). The card-based layout ensures a user-friendly experience, with different user roles quickly being able to access their distinct functions.

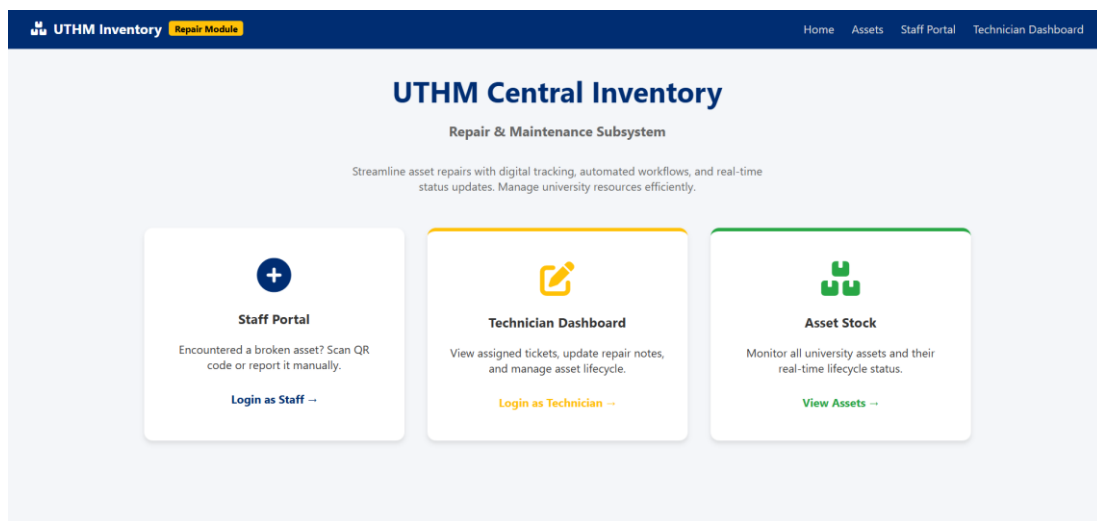


Figure 4.7.9.1 Landing Page

This figure shows the "Submit Defect Report" interface as part of the Staff Portal. Designed with efficiency in mind, it has a "Scan QR" button that can help staff instantly identify an asset without having to manually type anything. The form takes in important details such as a dropdown list to select the asset, a text area to detail the problem and an upload field for Photo Evidence. Submitting this form results in the system "Auto-Block" logic, where the status field of the asset is immediately updated to "Under Repair".

The screenshot displays the 'Submit Defect Report' form within the 'Staff Portal' of the 'UTHM Inventory Repair Module'. The interface includes a top navigation bar with links to 'Home', 'Assets', 'Staff Portal', and 'Technician Dashboard'. A '+ Report Issue' button is located in the top right corner of the portal area. The form itself is titled 'Submit Defect Report' and contains three main sections: 'Asset' with a dropdown menu labeled '-- Select Asset manually --' and a yellow 'Scan QR' button; 'Description of Issue' with a text area labeled 'Describe the problem in detail...'; and 'Photo Evidence' with a dashed border and a camera icon, accompanied by the text 'Click to upload or drag and drop'. A link to 'Back to Reports' is visible on the left side of the form.

Figure 4.7.9.2 Reporting Page

This dashboard helps to give an overview of the real-time asset inventory at the University. The upper section includes summary cards for keeping track of the number of assets in various states of the lifecycle: Available, Under Repair, and Disposed. The detailed table below enumerates specific assets with Serial Numbers, Locations and color coded Status Badges, allowing management to immediately see the status of the health and availability of campus equipment.

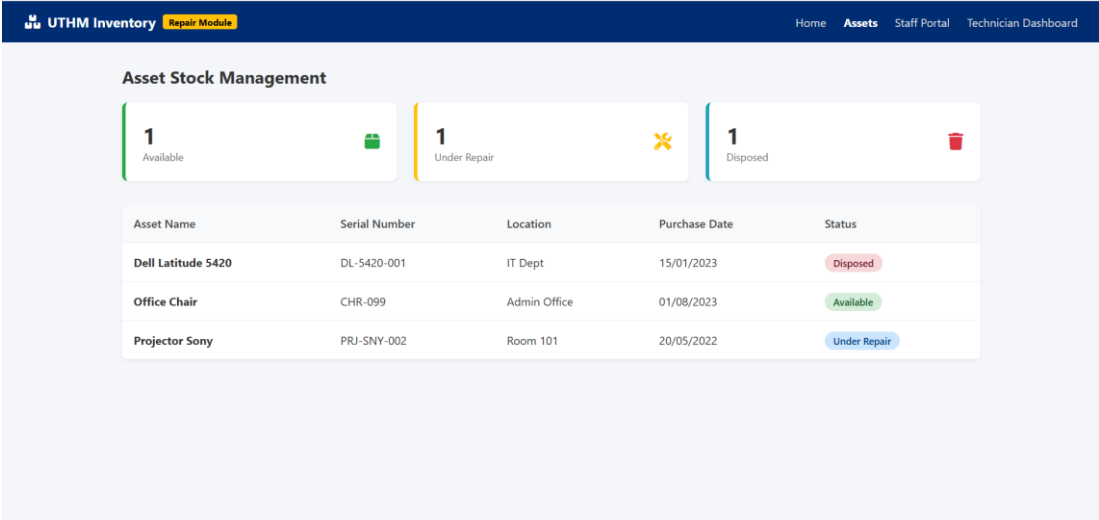


Figure 4.7.9.3 Asset Stock Management Dashboard

This figure illustrates the Repair Queue, the IT Technician's main working space. It has status indicators for Pending, In Progress, and Fixed tickets to help prioritize the workload. The interface has a list of active and previous maintenance tickets with Ticket ID, Asset details, and the Reporter. It enables technicians to trace how the resolution is (e.g. "Fixed" or "Unrepairable") which is directly updated in the main asset database using the system's backend synchronization logic.

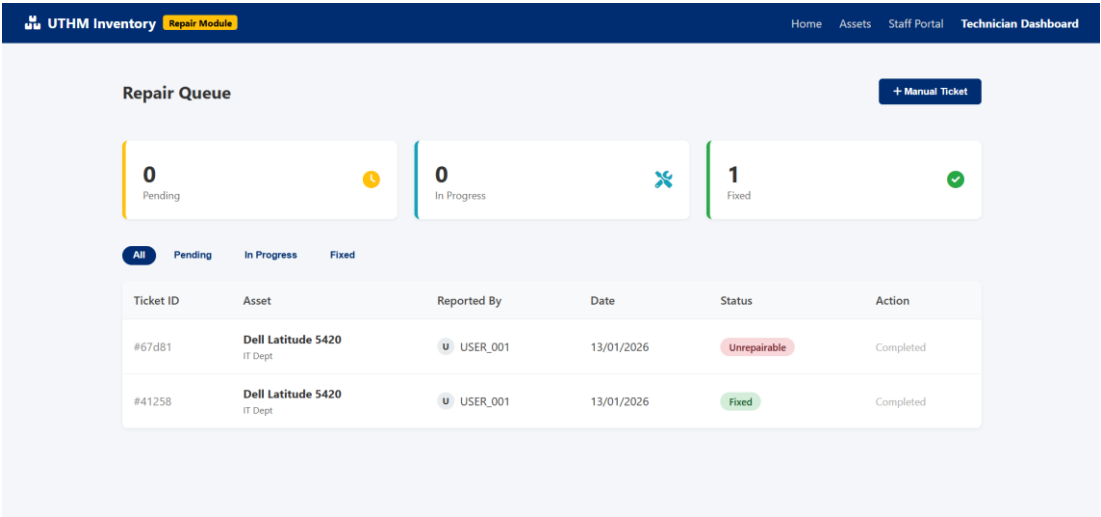


Figure 4.7.9.4 Technician Dashboard

4.7 Summary

To summarize, this chapter gave a critical analysis of the organizational hierarchy and the current As-Is architecture of UTHM. The synthesis of the requirements of the stakeholders collected during the structured interviews resulted in a To-Be Enterprise Architecture that proposed to create a single, cloud-based ecosystem with centralized data management and a Single Sign-On (SSO) integration. To ensure a seamless transition of the high-level architectural planning and technical implementation, the project has come up with certain design artifacts of the UTHM Inventory Subsystem, such as Use Case Diagrams and Use Case Descriptions to define the interactions of the actors, Activity Diagrams to show the procedural flows, and Sequence Diagrams to show the real time logic. These behavioral models, along with end-to-end database and interface designs by leveraging the SAP BTP Trial Account, The proposed system is also guarantee that it is structurally sound, user-focused, and well-equipped to take the development phase with the help of SAP BTP technology to facilitate the digital transformation of the university.

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 Introduction

This chapter documents the technical realization of the UTHM Inventory Subsystem (Repair Module). Following the design architecture in Chapter 4, the implementation phase transforms the architecture diagram to a functional software module. The primary purpose of this chapter is to show how the Repair Module was built and solve the challenges that track the broken asset this key problem. It details the database environment setup, backend logic implementation and allowing staff report issue and technicians to resolve them.

5.2 System Development

The system development followed the SAP BTP Cockpit workflow “Code-to-Cloud”. The data model was defined by the core data service (CDS), that service will auto deploy the physical schema to the SAP HANA Cloud database.

5.2.1 Create Database (Data Modeling)

Instead of manual SQL scripts, the database structure was defined in db/schema.cds file. The file define the Entities and their associations.

```
schema.cds U X
1 namespace uthm.inventory;
2
3 using { managed, cuid } from '@sap/cds/common';
4
5 // Entity 1: Assets (The Inventory Items)
6 entity Assets : managed, cuid {
7     name : String(100);
8     serial_number : String(50);
9     location : String(50);
10    // Status can be: 'Available', 'Under Repair', 'Disposed'
11    status : String(20) default 'Available';
12    purchase_date : Date;
13    repair_history : Association to many Repairlogs on repair_history.asset = $self;
14 }
15
16 // Entity 2: Repairlogs (The Maintenance Tickets)
17 entity Repairlogs : managed, cuid {
18    // Association links Repairlog directly to an Asset
19    asset : Association to Assets;
20    reporter_id : String(50); // User ID from SAP XSUAA
21    description : String(500);
22    damage_photo : LargeString; // Stores URL or Base64 string
23    // Status can be: 'Pending', 'Fixed', 'Unrepairable'
24    ticket_status : String(20) default 'Pending';
25    technician_note : String(500);
26 }
```

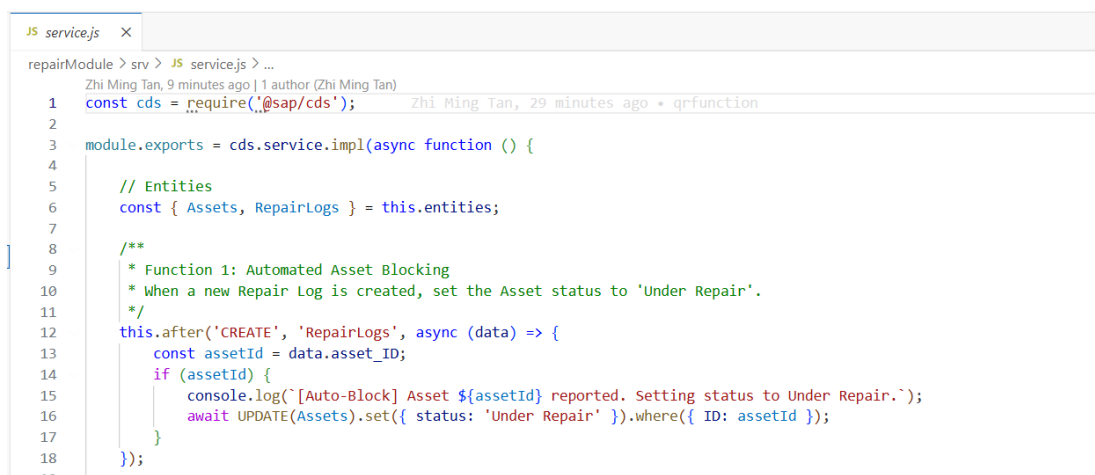
Figure 5.2.1.1 db/schema.cds

5.3 Coding of the System's Main Functions

The auto-blocking feature is implemented in the Service layer. In SAP CAP, we using the Event Handlers to intercept the database operation.

Function 1 : Automated Asset Blocking

When a staff create new repair log , the system must intercept this request and update the reported item status to Under Repair. This is to prevents the asset being borrowed while damaged.



```
JS service.js X
repairModule > srv > JS service.js > ...
Zhi Ming Tan, 9 minutes ago | 1 author (Zhi Ming Tan)
1 const cds = require('@sap/cds'); Zhi Ming Tan, 29 minutes ago • qfunction
2
3 module.exports = cds.service.impl(async function () {
4
5     // Entities
6     const { Assets, RepairLogs } = this.entities;
7
8     /**
9     * Function 1: Automated Asset Blocking
10    * When a new Repair Log is created, set the Asset status to 'Under Repair'.
11    */
12    this.after('CREATE', 'RepairLogs', async (data) => {
13        const assetId = data.asset_ID;
14        if (assetId) {
15            console.log(`[Auto-Block] Asset ${assetId} reported. Setting status to Under Repair.`);
16            await UPDATE(Assets).set({ status: 'Under Repair' }).where({ ID: assetId });
17        }
18    });
19
```

Figure 5.2.1.2 srv/service.js

Function 2 : Technician Resolution and Release

The system listens for an UPDATE event on RepairLogs entity and modify the ticket_status. If the status is marked as fixed, the linked asset will automatically reset to available, whereas a status of unrepairable trigger a permanent update to disposed.

```
21  * Function 2: Technician Resolution and Release
22  * When a Repair Log is updated (e.g. status change), update the Asset status accordingly.
23  */
24  this.after('UPDATE', 'RepairLogs', async (data) => {
25    if (data.ticket_status) {
26      let assetId = data.asset_ID;
27      if (!assetId) {
28        const log = await SELECT.one.from(RepairLogs).where({ ID: data.ID });
29        if (log) {
30          assetId = log.asset_ID;
31        }
32      }
33
34      if (assetId) {
35        let newAssetStatus = null;
36
37        if (data.ticket_status === 'Fixed') {
38          newAssetStatus = 'Available';
39        } else if (data.ticket_status === 'Unrepairable') {
40          newAssetStatus = 'Disposed';
41        } else if (data.ticket_status === 'In Progress') {
42          newAssetStatus = 'Under Repair';
43        }
44
45        if (newAssetStatus) {
46          console.log(`[Auto-Resolve] Ticket ${data.ID} became ${data.ticket_status}. Updating Asset ${assetId} to ${newAssetStatus}`);
47          await UPDATE(Assets).set({ status: newAssetStatus }).where({ ID: assetId });
48        }
49      } else {
50        console.log(`[Auto-Resolve] Warning: Could not find Asset ID for Ticket ${data.ID}`);
51      }
52    }
53  }
```

Figure 5.2.1.3 backend logic of Release

5.4 Summary

This chapter details the successful implementation of Repair Module from Inventory Subsystem by using enterprise-grade SAP BTP technology. The critical business logic was coded using SAP CAP Event Handlers to enable the auto-blocking which eliminates from the broken item being borrowed, and the resolution sync, which automatically return assets to stock or flag them as disposal base on technician input. The implementation concludes with the successful deployment of application service to the Cloud Foundry Runtime providing a secure, scalable backend support.

CHAPTER 6

CONCLUSION

6.1 Introduction

This final chapter provides a reflective summary of the UTHM Inventory System project. It outlines the primary achievements of the development process, acknowledges the limitations encountered during the implementation in the SAP Build Apps environment, and proposes strategic suggestions for future enhancements. The conclusion reinforces how the system fulfills the original objective of digitizing the manual inventory and borrowing processes at the university.

6.2 System Contribution/Achievement

The UTHM Inventory System has successfully introduced several key improvements to the management of university assets:

- **Centralized Data Management:** The transition from manual records to a structured relational database (On-Device Storage) ensures that user, item, and transaction data are synchronized and easily retrievable.
- **Automation of Approval Workflows:** By implementing sequential logic, the system automatically updates item availability statuses upon clerk approval, reducing administrative workload and preventing data entry errors.
- **Enhanced User Transparency:** Borrowers can now view real-time item availability and report damages directly through the mobile interface, fostering better accountability for university property.
- **Real-time Damage Reporting:** The integration of a dedicated damage reporting module allows the maintenance team to identify and isolate faulty equipment immediately.
- **Technical Skill Requirement:** The system requires personnel with adequate technical knowledge to maintain, configure and troubleshoot the application.

Limited expertise in SAP BTP and related technologies may slow down system enhancements and issue resolution.

- **Limited System Integration:** The current prototype operates as a standalone system and has limited integration with existing organizational systems. This may cause data error and reduce efficiency until proper system integration is implemented.

6.3 System Constraint

Despite the successful deployment of the prototype, several constraints were identified:

- **Offline Data Dependency:** As the system currently utilizes On-Device Storage, data is localized to the device. Real-time synchronization across multiple different clerk devices would require an external cloud database integration.
- **Cost Implementation:** By implementing SAP Business Technology Platform (BTP) environment requires a significant amount of investment and long-term commitment from the IT steering committee.
- **Technical Skill Requirement:** The system requires personnel with adequate technical knowledge to maintain, configure and troubleshoot the application. Limited expertise in SAP BTP and related technologies may slow down system enhancements and issue resolution.
- **Limited System Integration:** The current prototype operates as a standalone system and has limited integration with existing organizational systems. This may cause data error and reduce efficiency until proper system integration is implemented.

6.4 Future Suggestion

To further evolve the UTHM Inventory System, the following enhancements are recommended:

- **Cloud Backend Integration:** Transitioning the database to a cloud-based service that is also available in SAP BTP to allow real-time data sharing between students and multiple administrative departments.
- **QR Code/Barcode Integration:** Implementing a scanning feature using the device camera to allow users to scan an asset tag, which would automatically pull up the item's details and all the information users need to know.
- **Deep Integration with Financial Systems:** The system should further evolve to integrate directly with the Financial System via API. This would allow the automated calculation of asset depreciation and block the student account in LMS if they fail to return borrowed equipment on time.
- **Internet of Things (IoT) & RFID Tracking:** To automate the stock checking process, future iterations might upgrade from passive QR codes to active RFID tags. By installing the IoT sensor in laboratories, the system could auto detect the position of the high value asset , and no more manual scan of each asset by staff.
- **Predictive Analytics and Smart Dashboard Enhancement:** The system could incorporate predictive analytics powered by SAP analytics tools to forecast asset usage, maintenance cycles, and replacement needs. By analysing historical data from borrowing, repair, and disposal modules, the dashboard can provide insights such as high-demand items, frequently damaged assets, and upcoming procurement requirements. This would support proactive decision-making by administrators and reduce unexpected asset shortages.

6.5 Summary

In conclusion, one can say that the project concerning the deployment of the UTHM Inventory System has been able to demonstrate the efficacy of applying the low-code development to resolving the issue of the campus administration. The system provides a robust asset management system through making the process of borrowing and digitizing of inventory records simpler. Even though the data synchronization has some technical limitations, the recommendations provided in the future present a clear path of extending the application to become a full university wide.

6.6 Reflection

The literature review to the last implementation process had changed my outlook to being a data engineer rather than a solutions architect. At first, I thought that this project was just a straightforward project such as a regular system, but when I was informed of the legacy information silos of UTHM that the actual problem was something to do with the relationship of the structure, and not necessarily merely the functionality of the software alone. One of the most important moments was the designing of the Repair module. It compelled me to prove the fact of the "Auto-Blocking " workflow by a very strict diagramming prior to writing a single line of code, which demonstrated that solid architecture is the precondition of successful development. Chapter 5 implementation was the most technologically fulfilling because it served as a transition between the academic theory and enterprise reality. Xoding SAP CAP Event handler to achieve the automate asset governance was a steep learning curve, however, when the feature successfully deployed, it allowed me to realize how powerful is the cloud ecosystem ERP. Finally, the conclusion of the AI and IoT roadmap is that, as a digital transformation is a continuous process, I am now eager to be a strategic contributor to such a process.

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